

SmartTrans User Guide

A demonstration platform that uses AI-powered traffic accident analysis to make five foundational concepts of enterprise AI tangible and experiential.

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1 Executive Summary

SmartTrans illustrates five concepts that together define the modern AI agent stack:

Concept	The Question It Answers
Prompt	How do you give an AI a professional identity — so it says the <i>right</i> thing?
Agent	How do you combine role, knowledge, and tools into a digital specialist that completes tasks?
RAG	How do you give an AI access to authoritative reference documents — so it cites facts, not memories?
MCP	How do you equip an AI with tools — so it executes actions, not just offers advice?
Skills	How do you make AI expertise modular — so domain experts can evolve capabilities without engineering?

These five form a capability stack: **Prompts** define identity → **Agents** wrap identity, knowledge, and tools → **RAG** provides authoritative knowledge → **MCP** delivers executable tools → **Skills** make expertise swappable.

2 Core Concepts at a Glance

Concept	One-Line Understanding
Prompt	A “professional contract” — defines which path the AI <i>should</i> take
Agent	Role + knowledge + tools = a digital specialist that completes tasks
Structured Output	Verifiable, auditable data structures instead of free-form text
RAG	A reference library — look up facts rather than rely on training memory
Vector Embedding	Making semantics computable — turning meaning into measurable math
Tool Calling	AI autonomously decides when and which tool to invoke
MCP	A universal protocol for tool integration — one standard, any service
Skill	Modular expertise package — upgrade capabilities without code changes

3 Prompt — The Professional Contract

The core question: How do you make an AI go from “chatting” to “being professional”?

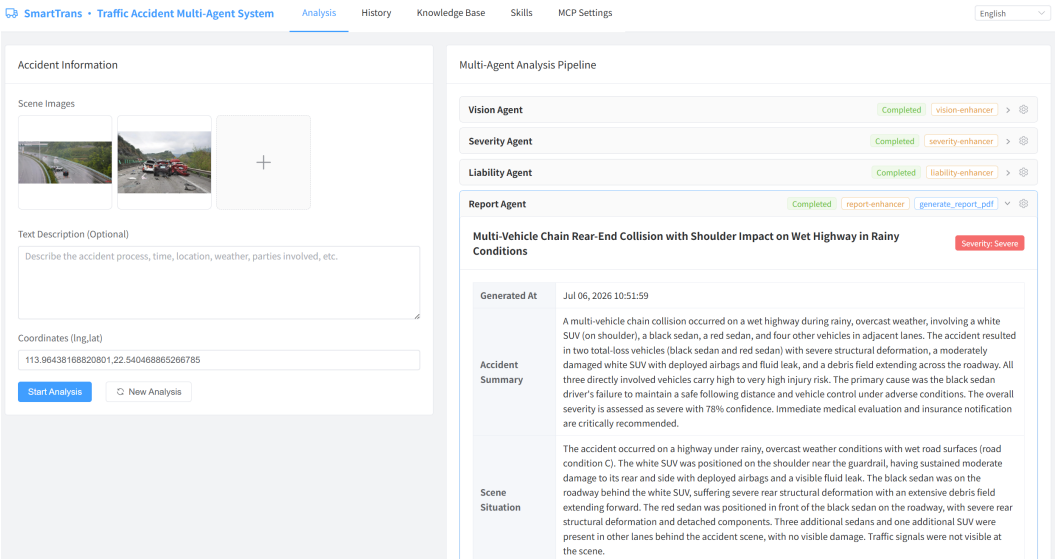
3.1 Experience the Pipeline

1. Open the system. Confirm you are on the **Accident Analysis** tab (denoted 「」 in the UI).
2. In the left panel **Case Information** area:
 - Click + to upload traffic accident scene images.
 - Fill in the **Text Description** field, for example:

At 3:00 PM on June 15, 2024, at a major urban intersection, Vehicle A traveling straight collided with Vehicle B making a left turn. Weather was clear, road surface was dry.

- Click **Start Analysis**.
3. Watch the pipeline execute in the right panel and wait for the report.

Tip: The language switcher in the top-right corner lets you choose English, Simplified Chinese, or Traditional Chinese. The entire pipeline operates in your selected language.



3.2 The Multi-Agent Architecture

The system decomposes analysis into a relay of four **specialized agents**:

Stage	Agent	Responsibility
	Vision Agent	Understand the visual information of the accident scene
	Severity Agent	Assess accident severity level and damage risks
	Liability Agent	Determine fault percentages and cite legal references
	Report Agent	Synthesize all analyses into a decision-support report

Status indicators update in real time: **Waiting** → **Analyzing** → **Complete**. Output of one stage is input to the next.

3.3 Why Prompts Matter

Three of the four agents (Severity, Liability, Report) are powered by the **same underlying AI model** (DeepSeek-V4). Yet they behave as entirely different specialists. The difference is entirely in their prompts.

Consider the same accident photo shown to different observers:

- Traffic officer → focuses on liability-relevant details.
- Insurance adjuster → focuses on loss-relevant details.
- Bystander → just sees “two cars crashed.”

Core Insight: A prompt is a “professional contract.” One model can serve dozens of business functions — each defined by its prompt, not its architecture.

3.4 Why Multi-Agent

A single AI handling everything leads to **attention dilution** — switching between vision, assessment, and legal reasoning. Multi-agent is **cognitive division of labor**: each agent has one responsibility, communicates through structured output, and deviations are localized.

3.5 Structured Output: The Dependable Interface

Every agent’s output is a **rigorously constrained data structure** — not free-form text. Structured data can be validated, transmitted, and audited. It is the dependable interface between agents.

```
[server] [2026-07-06 18:53:21] [INFO ] [orchestrator] Stage complete: severity
[server] {
[server]   "level": "severe",
[server]   "injuryRisk": "For the white SUV: High risk due to airbag deployment, fluid leak (may indicate occupant compartment breach), and side/rear impact; occupants likely require medical attention. For the black sedan (on roadway): Very high risk due to severe rear structural deformation and debris field, suggesting high-energy impact and possible entrapment or serious injuries. For the red sedan (in front): Very high risk due to severe rear structural deformation, high likelihood of whiplash, spinal injury, or head trauma. Vehicles in other lanes: Injury risk is unknown and assumed minimal unless involved in secondary impacts.",
[server]   "propertyDamage": "White SUV: Medium to Heavy damage (airbags deployed, fluid leak, moderate structural damage). Black sedan: Total loss (severe rear structural deformation, debris field). Red sedan: Total loss (severe rear structural deformation, detached components). The debris field and fluid leak also indicate probable damage to roadway infrastructure and environmental contamination, increasing overall property damage estimate well above $50,000.",
[server]   "confidence": 0.78,
```

4 Agent — The Digital Specialist

The **core question**: What exactly *is* an Agent, and how does it differ from a chatbot?

4.1 What Defines an Agent

Element	Provided By	What It Does
Role	Prompt	Defines professional identity, scope, and methodology
Knowledge	Training data + RAG	Access to facts — trained and retrievable
Tools	MCP	Ability to act — generate files, query services, call APIs

An agent is a **digital specialist**: defined scope + reference materials + ability to act. It executes tasks end-to-end, not just responds to prompts.

4.2 Agent vs. Chatbot

Dimension	Chatbot	Agent
Mode	Answers questions	Completes tasks
State	Stateless	Stateful — carries context through multi-step process
Output	Free-form text	Structured, validated data
Capability	Text generation only	Tool-equipped — retrieve, compute, generate, call APIs
Reasoning	Single-turn	Multi-step, autonomous tool-use decisions
Accountability	Output unverifiable	Output validated against schema

A chatbot tells you what you *should* do. An agent *does* it.

4.3 The Four Agents of SmartTrans

Agent	Model	Prompt Defines...	Tools Include...
Vision Agent	Qwen3-VL (vision)	How to observe accident scenes	— (pure perception)
Severity Agent	DeepSeek-V4 (reasoning)	How to classify damage and risk	RAG legal references
Liability Agent	DeepSeek-V4 (reasoning)	How to determine fault, cite law	RAG legal references
Report Agent	DeepSeek-V4 (reasoning)	How to synthesize into a report	PDF generation, map geocoding

The pipeline is a **data relay**, not a conversation.

Core Insight: An agent transforms a general-purpose model into a reliable, auditable, tool-equipped digital worker that plugs into business processes.

5 RAG — The Knowledge Foundation

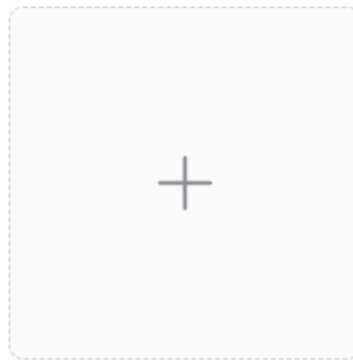
The core question: What gives an AI the authority to say “according to relevant laws”?

5.1 Experience It

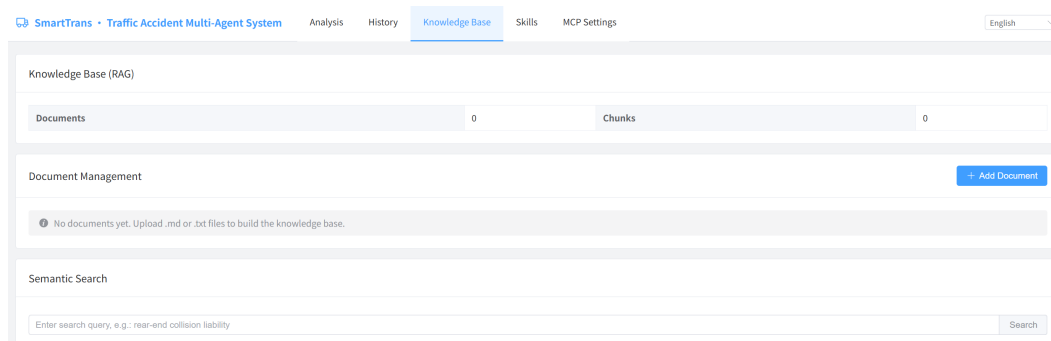
1. Switch to the **Knowledge Base** tab.
2. Upload documents: Click **Add Document** → drag a `.md` or `.txt` file → click **Upload**.

Accident Information

Scene Images



3. Try **semantic search**: enter **rear-end collision liability determination** → click **Search**. Observe **distance scores** — they measure semantic closeness.



4. Return to **Accident Analysis**, re-run, and compare Liability output — before and after having a knowledge base.

5.2 What Changed

Without a knowledge base: “according to relevant laws and regulations...” — vague, unverifiable.

With a knowledge base: specific articles cited, verbatim text shown, relevance to case explained.

Liability Agent

Completed

liability-enhancer

Party Responsibility

Black sedan

80%

The black sedan driver, on a wet road with rain, failed to maintain a safe following distance and control, causing a rear-end collision with the red sedan and then striking the white SUV on the shoulder. This conduct falls short of the standard expected of a competent and careful driver under the circumstances (road condition, weather, traffic), constituting dangerous driving as it created danger of injury or serious damage.

White SUV

15%

The white SUV was positioned on the shoulder near the guardrail, which may present an unexpected obstacle. If the driver stopped without hazard lights or in a non-emergency situation, that could amount to a failure to act as a competent driver. However, due to lack of details, only minor fault is assigned.

Red sedan

5%

The red sedan was ahead of the black sedan on the roadway. Without evidence of any improper action (e.g., sudden braking without cause), it bears minimal residual fault for possibly contributing to the following driver's reaction. The allocation is conservative given the lack of data.

Cited Legal Articles

Cap. 374 Road Traffic Ordinance Section (7)

For the purposes of subsections (4) and (5), in determining what would be expected of, or obvious to, a competent and careful driver in a particular case, regard shall be had to all the circumstances of the case including— (a) the nature, condition and use of the road concerned at the material time; (b) the amount of traffic which is actually on the road concerned at the material time or which might reasonably be expected to be on the road concerned at the material time; and (c) the circumstances (including the physical condition of the accused) of which the accused could be expected to be aware and any circumstances (including the physical condition of the accused) shown to have been within the knowledge of the accused.

Cap. 374 Road Traffic Ordinance Section (6)

For the purposes of subsections (4) and (5), ***dangerous*** refers to danger either of injury to any person or of serious damage to property.

Liability Conclusion

The scene evidence is insufficient to determine the exact sequence of events. Based on the positions and damage patterns, the most plausible scenario is that the black sedan (following too closely on a wet, rainy highway) first rear-ended the red sedan, then swerved and struck the white SUV stationary on the shoulder. The black sedan driver violated the duty of a competent and careful driver by failing to adjust speed and following distance for the adverse weather and road conditions. The white SUV driver may bear partial fault if the vehicle was stopped without adequate warning lights or in a prohibited location, but no evidence confirms this. The red sedan driver appears to have had no avoidable role. Given the uncertainty, fault is allocated as follows: black sedan 80% (primary cause of both collisions), white SUV 15% (potential contributory negligence for being stopped on shoulder without clear warning), red sedan 5% (residual possibility of sudden braking or lane position change, though unsubstantiated). The sum equals 100%.

5.3 The Essence of RAG

Rather than making the AI memorize everything, give it a reference library. Three problems RAG solves: knowledge cutoff (documents update anytime), hallucination risk (constrained to real documents), domain depth (specialist knowledge without retraining).

Core Insight: RAG means your proprietary documents become the AI's source of truth — updatable, auditable, controlled by you.

5.4 Chunking and Vectorization

- **Chunking:** The system detects **Article X** markers and splits by article. Precise retrieval requires precise granularity.
- **Vectorization:** Each unit → 4,096-dimensional vector. “Rear-end collision” and “vehicle behind strikes rear of vehicle ahead” are different in characters but close in vector space — vectors capture *meaning*, not *spelling*.

6 MCP — From Advisor to Executor

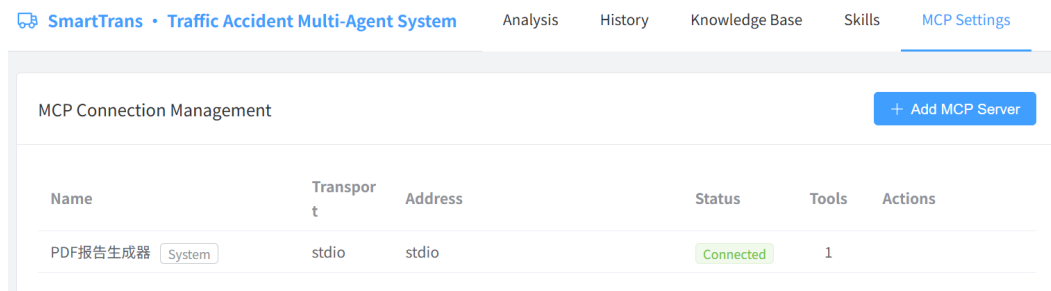
The core question: Can an AI do more than “talk” — can it actually “act”?

6.1 Prerequisites

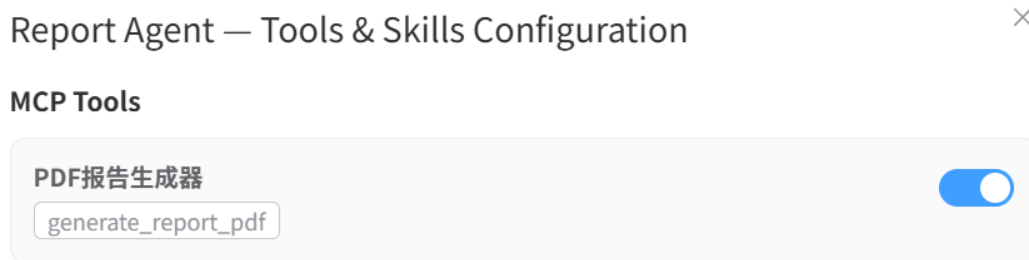
MCP must be enabled server-side (`MCP_ENABLED=true`). If the **MCP Settings** tab is not visible, contact the administrator.

6.2 Experience It

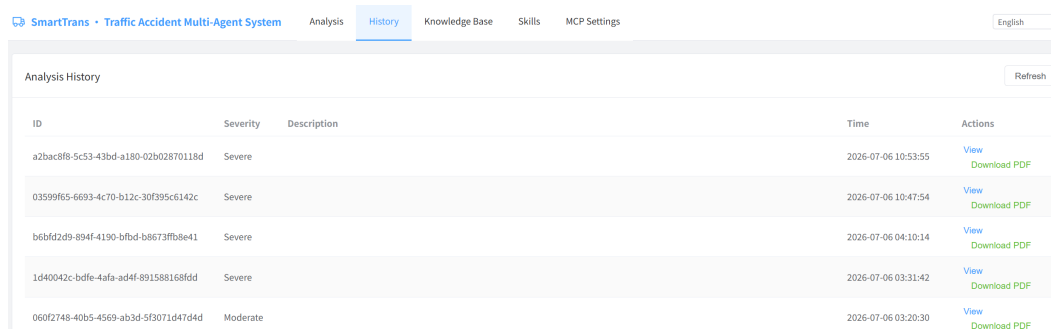
1. Switch to **MCP Settings**. You will see a System connection: **PDF Report Generator** — a preset that generates formal PDF documents.



2. Enable the tool: go to **Accident Analysis** → click the gear icon on the **Report Agent** → **MCP Tools** tab → toggle **PDF Report Generator** on.



3. Run analysis → switch to **History** → click **Download PDF** to get a formatted report.



6.3 Tool Calling: Crossing from Advice to Action

Understand the need → decide which tool to call
→ pass parameters → receive results → continue reasoning

The AI itself judges *when* to call, *which* tool, and *how* to interpret the result — not a pre-scripted workflow.

Core Insight: Without tools, AI is a “consultant.” With tools, AI becomes an “executor” — crossing from decision support to process automation.

6.4 MCP: A Universal Protocol

MCP solves the N×M scaling problem (10 systems × 50 APIs = 500 custom integrations). One standard for discovery, invocation, and extension. MCP is to AI tools what USB is to computer peripherals.

6.5 System Connections vs. User Connections

Type	Characteristics	Management
System Connections	Preset, non-deletable; platform-native capabilities	Platform maintains
User Connections	User-added; on-demand external capabilities	User manages lifecycle

Use **Add MCP Server** to connect external services — mapping, weather, databases — growing the AI’s capability map organically.

7 Skills — Evolve Without Rewriting Code

The core question: Can we upgrade an AI agent’s capabilities without touching code?

7.1 Experience It

- Switch to the **Skills** tab. Four pre-installed system skills, each System and non-deletable:

Skill	Default Agent	Purpose
vision-enhancer	Vision	Damage classification, road condition grading, weather/lighting analysis
severity-enhancer	Severity	Injury risk classification, multi-factor severity matrix
liability-enhancer	Liability	Fault assessment criteria, accident-type responsibility splits
report-enhancer	Report	Report template, post-accident checklists, language consistency

- Create a custom skill: Click **New Skill** → paste a **SKILL.md** document:

```
---
name: severity-checklist
description: |
  A structured checklist for comprehensive severity assessment,
  covering vehicle damage grades, injury classification,
  and environmental hazard evaluation.
---
```

```
# Severity Assessment Checklist
```

```
## Instructions
```

```
When assessing accident severity, systematically evaluate:
```

```
### 1. Vehicle Damage
```


- Level A: Cosmetic damage only (scratches, dents)
- Level B: Functional damage (lights, mirrors, windows)
- Level C: Structural damage (frame, axles, crumple zones)
- Level D: Total loss

2. Injury Classification

- Minor: Bruises, whiplash – outpatient treatment
- Moderate: Fractures, lacerations – hospitalization < 7 days
- Severe: Life-threatening, permanent disability – hospitalization ≥ 7 days
- Fatal: One or more fatalities

3. Environmental Hazards

- Fluid spills (fuel, oil, coolant)
- Road blockage (partial or full lane closure)
- Secondary collision risk (blind curves, high-speed traffic)

3. Bind to agent: **Accident Analysis** → gear icon on **Severity Agent** → **Skills** tab → toggle **severity-checklist** on.
4. Run analysis — the Severity Agent step shows a severity-checklist tag and the assessment reflects the structured checklist approach.

7.2 What Happened

A **Skill** is a reusable capability package stored as **SKILL.md** under **server/data/skills/<skill-name>/**. At runtime:

1. **SkillsManager** parses all **SKILL.md** files at startup.
2. The orchestrator queries enabled skills per agent (merging persisted bindings + per-request selections).
3. Skills are injected into the agent's system prompt with boundary markers:

```
--- BEGIN SKILLS ---
[Skill: severity-checklist]
Description: A structured checklist for comprehensive severity assessment...
Instructions:
# Severity Assessment Checklist
...
[/Skill: severity-checklist]
--- END SKILLS ---
```

7.3 Skills in the Capability Stack

Mechanism	Metaphor	What It Does
Prompt	Job description	Defines <i>what kind of expert</i> the agent is
RAG	Reference library	<i>What facts</i> the agent can look up
MCP / Tools	Hands and instruments	<i>What the agent can do</i> in the world
Skills	Training manual	<i>How the agent should think and operate</i>

Skills are **operational know-how** — heuristics, methodologies, checklists. They decouple capability upgrades from code changes. A domain expert authors a **SKILL.md**; no deployment, no retraining, no pipeline reconfiguration.

Core Insight: Prompts define *what the agent is*. RAG provides *what it can reference*. MCP enables *what it can do*. Skills define *how it thinks* — modular, swappable, authorable by domain experts.

7.4 The SKILL.md Format

```

---
name: <unique-identifier>
description: |
  <what this skill provides – used for discovery and selection>
---

```

```

# <Title>

## Instructions

<Guidance, heuristics, checklists, or methodologies.
Use standard Markdown – headings, tables, lists.>

```

The **name** is the unique identifier. The **description** appears in the UI. The body is injected into the agent’s system prompt — write it like a training manual for a new employee.

8 Summary: The Five-Layer Capability Stack

Layer	Concept	The Leap	Business Impact
1	Prompt	From “can talk” to “says the right thing”	One model serves many business functions
2	Agent	From “a model” to “a digital specialist”	Auditable, tool-equipped unit for business processes
3	RAG	From “remembering” to “looking it up”	Proprietary documents become the AI’s source of truth
4	MCP	From “advising” to “executing”	Enterprise services become AI-callable through one standard
5	Skills	From “built once” to “continuously evolving”	Domain experts upgrade AI without engineering